

Multiplication

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Multiplication is **repeated equal groups**. 3×4 means 3 groups of 4, which is 12. An **array** shows the same idea in rows and columns. For two-digit numbers, use the **area model**: break each number by place value and multiply the parts. $12 \times 13 = (10 + 2) \times (10 + 3) = 100 + 30 + 20 + 6 = 156$. The **standard algorithm** multiplies each digit and adds the partial products.

Round 1 — Retrieve It

1. What does 3×4 mean in words (groups and size)?

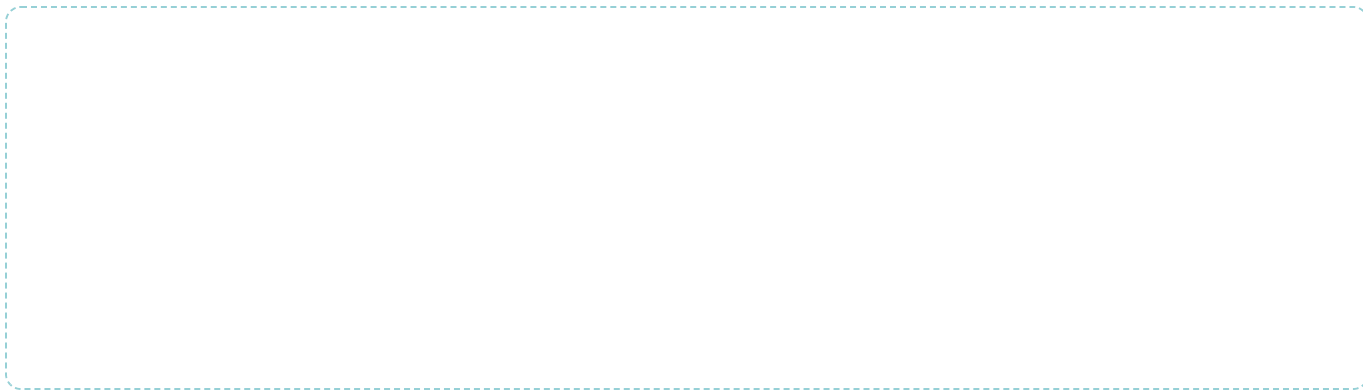
2. What is 3×4 ?

3. In an array, what do the rows and columns show?

4. In the area model, how do you break apart 12 to multiply it?

Round 2 — Sketch It

Draw an array that shows 3×4 . Use dots in 3 rows with 4 in each row, then write the total.



Multiplication — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. There are 6 bags with 5 apples in each bag. Write a multiplication sentence and find the total.

2. Use the area model to solve 12×13 . Show the four parts you multiply.

3. Solve 14×6 with the standard algorithm. Show the carry.

Round 4 — Reason from Evidence

A student solved 12×13 and got 36 by only doing 12×3 . Explain what parts they left out and give the correct product.

Multiplication — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how repeated groups and an array show the same multiplication. Use the words groups, rows, and total.

C**Connect**

Connect the area model to finding the area of a rectangle you have seen, like a floor or a garden. How are they alike?

E**Extend**

Use the area model to solve 15×12 . Show all four parts and add them to get the product.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Multiplication

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. 3 groups of 4 (three groups with four in each group).
2. 12.
3. Rows and columns show equal groups; the array shows the total as rows $\times$ columns.
4. Break 12 into $10 + 2$ (by place value).

Sketch It — look for

An array with 3 rows of 4 dots each (or 4 rows of 3), and the total written as 12.

Apply It

1. $6 \times 5 = 30$ apples.
2. $12 \times 13 = 156$. Parts: $10 \times 10 = 100$, $10 \times 3 = 30$, $2 \times 10 = 20$, $2 \times 3 = 6$; $100 + 30 + 20 + 6 = 156$.
3. $14 \times 6 = 84$ ($6 \times 4 = 24$, write 4 carry 2; $6 \times 1 = 6$, plus 2 = 8).

Reason from Evidence — look for

They left out 12×10 (which is 120). The full product is $12 \times 13 = (12 \times 10) + (12 \times 3) = 120 + 36 = 156$.

ACE — Extend look-for

$15 \times 12 = 180$. Parts: $10 \times 10 = 100$, $10 \times 2 = 20$, $5 \times 10 = 50$, $5 \times 2 = 10$; $100 + 20 + 50 + 10 = 180$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.